

ERC PROOF OF CONCEPT

European Research Council Proof of Concept

Commercialising a Selective CO₂-to-Ethylene Electrocatalyst CATALYX-PoC

Field	Value
Call Identifier	ERC-2026-POC
Type of Action	ERC Proof of Concept
Duration	18 months
TRL Range	TRL 3 to TRL 5
Total Budget	EUR 150,000
EU Contribution	EUR 150,000

Abstract

CATALYX-PoC exploits a breakthrough copper-based electrocatalyst - developed under an ERC Consolidator Grant - that converts CO₂ to ethylene with a Faradaic efficiency exceeding 85% and demonstrated long-term operational stability, outperforming all known competing materials on the combined metrics of selectivity, durability, and synthesis scalability. Ethylene is the world's highest-volume petrochemical, produced almost entirely from fossil feedstocks via a process responsible for approximately 200 Mt CO₂ per year; a selective electrochemical CO₂ conversion route would fundamentally disrupt this value chain and reduce industrial CO₂ emissions at scale. Over 18 months, with EUR 150,000 in ERC PoC funding, CATALYX-PoC will: (i) complete a full techno-economic assessment by month 9, quantifying commercial viability and identifying the cost-reduction levers critical to competitiveness with steam cracking; (ii) file a provisional patent application by month 3 and a PCT application by month 12, securing IP protection for the catalyst composition, synthesis method, and electrode architecture; (iii) validate prototype electrolyser performance at industrially relevant scale - 10 cm² active electrode area, Faradaic efficiency >80%, 500-hour continuous stability - by month 15; and (iv) engage at least six qualified industrial partners and document a committed exploitation pathway (licensing, spin-out, or joint development agreement) by month 15. The project will advance the technology from TRL 3 to TRL 5 and deliver a detailed commercialisation plan, establishing CATALYX as a leading candidate technology for decarbonising the global chemical industry and contributing directly to European Green Deal climate and industrial competitiveness objectives.

Keywords: CO₂ electroreduction, electrocatalysis, ethylene, copper catalyst, carbon capture and utilisation, green chemistry, Proof of Concept, commercialisation, IP strategy, techno-economic assessment

SECTION 1 - EXCELLENCE AND INNOVATION POTENTIAL

1.1 The Innovation and its Objectives

The ERC Consolidator Grant underpinning this Proof of Concept produced a copper-based electrocatalyst with an unprecedented combination of Faradaic efficiency (>85%) and selectivity towards ethylene in the electrochemical CO₂ reduction reaction (CO₂RR). This advance was achieved through atomic-level engineering of the copper surface coordination environment, suppressing competing pathways to methane, ethanol, and formate while stabilising the C–C coupling intermediate that directs selectivity to ethylene. The result constitutes a scientifically validated breakthrough at TRL 3, demonstrated under controlled laboratory conditions with fully reproducible performance metrics.

The innovation potential is both compelling and timely. Ethylene is the world's highest-volume petrochemical, with global production exceeding 200 million tonnes per year and an almost total dependence on steam cracking of fossil-derived naphtha or ethane - a process responsible for approximately 200 Mt CO₂ per year. A selective, electricity-driven CO₂-to-ethylene route would disrupt this value chain, enabling chemical producers to valorise captured CO₂ using renewable electricity and to deliver a drop-in feedstock with a deeply reduced carbon footprint. CATALYX-PoC directly exploits this ERC research output to determine whether the scientific breakthrough can be translated into a commercially viable technology, advancing it from TRL 3 to TRL 5 within 18 months.

The project pursues four SMART objectives: (O1) complete a full techno-economic assessment (TEA) identifying cost-competitiveness thresholds by month 9; (O2) file a provisional patent application by month 3 and a PCT application by month 12, securing IP protection for the catalyst composition, synthesis method, and electrode architecture; (O3) validate prototype electrolyser performance at 10 cm² electrode scale with Faradaic efficiency >80% and 500-hour operational stability by month 15; and (O4) engage at least six qualified industrial partners and document a committed exploitation pathway - licensing, spin-out, or joint development agreement - by month 15.

1.2 Link to the ERC Frontier Research Result

The ERC Proof of Concept instrument is designed precisely for the situation addressed by CATALYX-PoC: a well-defined, high-impact research result generated within an ERC frontier research grant that harbours clear commercial and societal potential but requires dedicated de-risking activities before industry will invest. The activities proposed here - techno-economic assessment, IP protection, prototype validation, and industrial partner engagement - draw directly and exclusively on the copper electrocatalyst innovation produced under the parent ERC Consolidator Grant. No new fundamental research is proposed, in full accordance with the ERC PoC eligibility requirements.

CATALYX-PoC also aligns with the European Green Deal, the EU Industrial Strategy, and the objectives of the Net-Zero Industry Act, all of which call for the rapid scale-up of carbon capture and utilisation (CCU) technologies. Electrocatalytic CO₂ conversion to value-added chemicals is explicitly identified as a priority CCU pathway in these policy frameworks. By advancing the catalyst towards commercial readiness, CATALYX-PoC contributes directly to European industrial competitiveness, EU climate targets for 2030 and 2050, and the strategic goal of establishing European leadership in green chemistry and electrochemical manufacturing.

1.3 Technical Approach and Proof-of-Concept Plan

The PoC programme is structured around four integrated and sequentially gated activity streams, each designed to generate decision-relevant outputs that inform subsequent activities.

First, a rigorous techno-economic assessment (TEA) will quantify the cost of ethylene produced via the CATALYX electrocatalyst under realistic operating assumptions, identify the dominant cost drivers

(electricity price, catalyst lifetime, membrane performance, and stack engineering), and map the conditions under which the technology achieves cost-competitiveness with incumbent steam cracking. The TEA will directly inform go/no-go decisions and shape the IP scope and partnership strategy.

Second, the PI will work with the University of Cambridge's technology transfer office (Cambridge Enterprise) to conduct a freedom-to-operate (FTO) analysis and file priority patent applications covering the catalyst composition, synthesis method, and electrode architecture. This work begins in month 1 to ensure that prototype activities and external partner communications remain within a legally protected boundary.

Third, a prototype electrolyser cell will be assembled and operated at the 10 cm² electrode scale to generate the performance data required by industrial partners for independent evaluation - specifically Faradaic efficiency, partial current density, and stability over 500 hours of continuous operation under industrially representative conditions.

Fourth, targeted engagement with chemical producers, electrolyser manufacturers, and venture-backed CCU start-ups will be conducted to identify the most viable route to commercial deployment, whether through licensing, spin-out formation, or a joint development agreement.

The methodology is deliberately milestone-driven: TEA findings gate the patent filing scope; prototype data gate the partner engagement phase; partner feedback gates the exploitation pathway decision. This sequencing ensures that the EUR 150,000 budget is deployed with maximum strategic leverage and that each activity generates actionable intelligence for the next. Open Science practices are observed throughout: scientific results that do not compromise patentability will be published in peer-reviewed journals under open access, in accordance with the ERC mandate. Gender-disaggregated data will be collected for all personnel recruited to the project, and equal opportunity principles will be applied in all hiring decisions.

1.4 Interdisciplinary Considerations

The novelty of the CATALYX electrocatalyst relative to existing CO₂RR technologies rests on three mutually reinforcing and commercially decisive distinguishing features.

First, selectivity: while state-of-the-art copper catalysts reported in the literature achieve Faradaic efficiencies for ethylene of 50–70%, the CATALYX material consistently exceeds 85% under identical test conditions. This step-change materially reduces the energy and capital cost of downstream product separation, which is one of the principal economic barriers to commercial CO₂RR deployment.

Second, stability: through the surface coordination engineering strategy developed during the ERC project, the catalyst retains more than 90% of its initial activity after 200 hours of continuous operation, compared with typical literature degradation of 30–50% over the same period. Long-term stability is a prerequisite for industrial deployment and is routinely cited by industrial partners as the primary technical concern in CO₂RR electrocatalysis.

Third, synthesis scalability: the preparation route is based on wet-chemistry methods that are compatible with roll-to-roll electrode manufacturing, unlike the vapour-deposition approaches employed in competing high-performance catalysts, which are inherently difficult to scale.

No commercially available or near-market CO₂-to-ethylene electrocatalyst combines these three attributes. The closest competitive technologies are either more selective but operationally unstable, or sufficiently stable but with insufficient selectivity to be economically viable. CATALYX-PoC

therefore occupies a genuinely differentiated and defensible position in the innovation landscape. The PoC activities draw on expertise spanning electrochemistry, materials science, chemical engineering (for the TEA and process modelling), and technology commercialisation - an inherently interdisciplinary undertaking in which Cambridge Enterprise's IP and business development capabilities complement the PI's deep technical knowledge. If the PoC activities confirm commercial feasibility, CATALYX is positioned to define a new product category in the electrochemical CCU sector.

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SECTION 2 - IMPACT AND COMMERCIALISATION

2.1 Innovation Potential and Market Opportunity

The primary target market is the global ethylene production industry, valued at approximately USD 180 billion annually. The initial addressable segment is industrial CCU: chemical companies and energy majors investing in carbon capture and utilisation under mounting regulatory pressure from the EU Emissions Trading System and equivalent carbon pricing mechanisms in major economies. Within this segment, the most tractable early customers are integrated chemical producers operating large CO₂ point sources (crackers, refineries, and cement plants) that seek to convert captured CO₂ into saleable ethylene without requiring a complete process redesign. A secondary, near-term segment is the growing market for electrolyser systems and green-chemistry solutions supplied by equipment manufacturers and engineering companies seeking differentiated CO₂RR electrode technology.

Three candidate routes to market will be rigorously assessed and confirmed during the PoC period: (i) technology licensing to an established electrocatalyst or electrolyser manufacturer, leveraging Cambridge Enterprise's proven licensing capabilities and established industry relationships; (ii) formation of a University spin-out company to develop the catalyst into a commercial electrode product, with Cambridge Enterprise providing company formation support and access to the Cambridge entrepreneurial ecosystem; (iii) a joint development agreement with a chemical-industry partner that provides scale-up infrastructure and market access in exchange for co-development rights. The TEA results (available by month 9) and prototype validation data (available by month 15) will together provide the quantitative evidence base required to select among these pathways with confidence. A formal exploitation pathway decision, co-signed by the PI and Cambridge Enterprise, will be documented by month 15.

Societal impact is substantial and directly measurable. Commercial deployment of the CATALYX technology at scale would reduce the carbon intensity of ethylene production - one of the largest single-process industrial CO₂ sources globally - and would do so using renewable electricity and captured CO₂, contributing directly to EU 2030 and 2050 climate targets. Even a 5% penetration of the European ethylene market would avoid several million tonnes of CO₂ per year, consistent with the scale of impact required by the European Green Deal industrial transformation agenda.

2.2 Commercialisation and IP Strategy

IP protection is the foundational exploitation activity and is initiated on day one of the project. A provisional patent application will be filed in month 3, covering the catalyst composition, the coordination-engineering synthesis method, and the electrode architecture. A PCT application will follow in month 12, informed by the FTO analysis results and strengthened by the prototype validation dataset generated during WP3. Cambridge Enterprise will manage the filing process and oversee the FTO analysis, engaging specialist external patent attorneys with expertise in electrochemical materials. The IP strategy is designed to create a defensible position that supports either a licensing transaction or a spin-out company valuation, while remaining fully compliant with ERC grant agreement requirements on open access and the Granting Authority's rights regarding exclusive licensing.

Beyond formal IP, the PoC will generate three categories of high-value proprietary knowledge: a validated prototype performance dataset (500-hour continuous operation at 10 cm² scale), a full TEA report with sensitivity analysis, and a structured market analysis including qualified industrial partner feedback. These assets collectively constitute the commercial dossier required to support investment or partnership decisions.

Scientific results that do not compromise patentability will be published in peer-reviewed open-access journals in strict accordance with the ERC open access mandate, ensuring that the underlying science reaches the broader research community. The PI will present findings at two major industry-facing events - specifically CO₂ electroreduction and industrial CCU conferences - to build visibility with potential partners and establish CATALYX as a credible candidate technology. A targeted, professionally produced briefing document will be prepared for distribution to pre-qualified industrial contacts under non-disclosure agreements, serving as the primary instrument for structured partner engagement during WP4.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Faradaic efficiency for ethylene at prototype scale	>80% at 10 cm ² electrode over 500 h	Chronoamperometry with online GC product analysis
Patent applications filed	1 provisional (month 3) + 1 PCT (month 12)	Filing receipts from patent office
Industrial partner meetings	At least 6 qualified industrial contacts engaged	Signed NDAs and meeting records
Techno-economic assessment completed	Full TEA report with sensitivity analysis by month 9	Delivery of TEA report
Exploitation pathway decision	Documented go/no-go decision on licensing vs spin-out by month 15	Board-level decision memo with Cambridge Enterprise

SECTION 3 - IMPLEMENTATION

3.1 Activities and Timeline

WP	Title	Lead	PM	Duration
WP1	Techno-Economic Assessment	University of Cambridge	4	M1-M9
WP2	IP Protection and Freedom-to-Operate Analysis	University of Cambridge	2	M1-M12
WP3	Prototype Validation	University of Cambridge	6	M3-M15
WP4	Industrial Partner Engagement and Exploitation Pathway	University of Cambridge	4	M6-M18

WP1: Techno-Economic Assessment

Lead: University of Cambridge | Person-months: 4 | Duration: M1 - M9

WP1 develops a full techno-economic model for CO₂-to-ethylene production using the CATALYX electrocatalyst. The model encompasses capital cost estimation, operating cost analysis (with explicit sensitivity to electricity price and catalyst lifetime), mass and energy balances for an electrolyser-based process at 1 kt/year ethylene capacity, and systematic benchmarking against incumbent steam cracking and competing CCU pathways. The TEA will identify threshold performance parameters - minimum Faradaic efficiency, current density, and catalyst lifetime - that must be achieved for the technology to reach cost-competitiveness, thereby providing a technically grounded basis for the IP filing scope, the go/no-go decision framework, and the partner engagement narrative. Delivery of the completed TEA report by month 9 constitutes a critical gate for WP4 activities.

T1.1: Process modelling and mass balance - Construct a process flow model for a 1 kt/year ethylene electrolyser plant, encompassing CO₂ supply and purification, electrolyser stack, gas-liquid separation, and downstream ethylene purification. Mass and energy balances will be validated against published electrolyser data and the PI's own laboratory measurements.

T1.2: Cost modelling and sensitivity analysis - Parameterise capital expenditure and operating costs using engineering cost estimation methods. Execute a Monte Carlo sensitivity analysis across key variables - electricity price (EUR 30–120/MWh), catalyst lifetime (500–5,000 hours), Faradaic efficiency (70–90%), and stack cost - to identify minimum performance targets for cost-competitiveness and to quantify the value of further R&D improvements.

T1.3: Market benchmarking report - Compare the CATALYX process economics against steam cracking and the leading alternative CCU pathways (e.g., CO₂-to-methanol, CO₂-to-formic acid). Identify the optimal market entry point, target customer segment, and the regulatory and carbon pricing conditions under which the technology becomes investable. This report will feed directly into the partner engagement briefing document developed in WP4.

WP2: IP Protection and Freedom-to-Operate Analysis

Lead: University of Cambridge | Person-months: 2 | Duration: M1 - M12

WP2 establishes and defends the IP position required to support commercial exploitation of the CATALYX electrocatalyst. Working from day one in collaboration with Cambridge Enterprise and external patent counsel, the work package conducts a comprehensive FTO analysis of the CO₂RR electrocatalyst patent landscape, identifies available claim space, and prepares and submits patent applications with a scope tailored to both the scientific novelty and the commercial value of the innovation. The FTO analysis is deliberately front-loaded to ensure that all prototype activities and external partner communications in subsequent work packages take place within a legally protected boundary. IP activities are designed to be fully compliant with ERC grant agreement obligations, including open access requirements and the Granting Authority's rights.

T2.1: FTO landscape analysis - Commission and critically review a professional FTO analysis - conducted by external patent attorneys with expertise in electrochemical materials - covering all relevant claim categories: catalyst composition and morphology, coordination-engineering synthesis methods, electrode architecture, and electrolyser operating conditions. The analysis will identify freedom-to-file positions and flag any blocking prior art requiring design-around strategies.

T2.2: Provisional patent filing - Draft and file a provisional patent application in month 3, covering the core catalyst

innovation in the broadest claims supported by the ERC project data and the FTO findings. The provisional filing establishes priority date and provides a 12-month window for claim refinement based on prototype validation results.

T2.3: PCT application - Prepare and file a PCT application by month 12, incorporating the 500-hour prototype stability dataset from WP3 and extending or refining claim scope based on final FTO conclusions. The PCT filing will provide international protection in jurisdictions of commercial relevance, including the EU, USA, Japan, and South Korea.

WP3: Prototype Validation

Lead: University of Cambridge | Person-months: 6 | Duration: M3 - M15

WP3 advances the CATALYX electrocatalyst from laboratory-scale coupon testing (TRL 3) to a validated prototype electrolyser at the 10 cm² active electrode area (TRL 5). The work package generates the commercially relevant performance dataset - Faradaic efficiency, partial current density for ethylene, and 500-hour operational stability - under industrially representative conditions (gas-diffusion electrode configuration, alkaline electrolyte, continuous CO₂ feed) that industrial partners require to make independent technical assessments. The prototype dataset constitutes the primary technical evidence base for both the PCT patent filing (WP2) and the partner engagement activities (WP4). All experimental data are recorded in indexed laboratory notebooks and managed in accordance with the University of Cambridge's research data management policy, supporting open access publication of non-proprietary results.

T3.1: Electrode scale-up fabrication - Develop and optimise gas-diffusion electrode fabrication at 10 cm² scale using the CATALYX catalyst, systematically varying catalyst ink formulation, binder content, solvent system, and spray-deposition parameters. Performance at each fabrication condition will be benchmarked against the laboratory standard to identify scale-up losses and their origins.

T3.2: Electrolyser cell assembly and testing - Assemble a flow-cell electrolyser incorporating the scaled gas-diffusion electrode, perform polarisation curve measurements, and validate Faradaic efficiency and partial current density against laboratory-scale results. Operating conditions (electrolyte concentration, CO₂ flow rate, applied potential) will be systematically mapped to establish the performance envelope for the 500-hour stability run.

T3.3: 500-hour stability run - Operate the prototype electrolyser continuously for 500 hours under the optimised conditions identified in T3.2. Faradaic efficiency, current density, and cell voltage will be monitored in real time using online gas chromatography and electrochemical impedance spectroscopy. The resulting dataset - including degradation trajectory analysis - will be compiled into a structured partner disclosure document suitable for distribution under NDA.

WP4: Industrial Partner Engagement and Exploitation Pathway

Lead: University of Cambridge | Person-months: 4 | Duration: M6 - M18

WP4 translates the technical and economic evidence generated in WP1–3 into a committed commercial exploitation pathway for the CATALYX technology. The work package systematically identifies and engages qualified industrial partners across the chemical production, electrolyser manufacturing, and CCU investment sectors; conducts structured NDA-protected technical briefings supported by the prototype dataset and TEA report; and evaluates three candidate exploitation pathways - technology licensing, spin-out formation, and joint development agreement - on the basis of partner feedback and commercial attractiveness. The work package culminates in a formally documented exploitation pathway decision by month 15 and a detailed commercialisation plan submitted as part of the PoC final report by month 18. Cambridge Enterprise is actively involved throughout, providing commercial negotiation support and institutional continuity beyond the PoC grant period.

T4.1: Partner identification and outreach - Conduct a structured mapping of the industrial partner landscape, covering large integrated chemical producers (ethylene value chain), electrolyser original equipment manufacturers, engineering and technology licensing companies, and CCU-focused venture investors. Initiate contact with at least ten qualified prospects, prioritising organisations with existing CCU programmes and demonstrated capacity to scale electrochemical processes. The PI's existing industry network from the parent ERC project and Cambridge Enterprise's sector contacts will be leveraged to ensure warm introductions where possible.

T4.2: Partner meetings and technical briefings - Conduct in-depth NDA-protected technical briefings with at least six qualified industrial partners, presenting the prototype performance dataset, TEA findings, and IP position. Structured feedback will be collected on technical requirements, commercial terms, and preferred partnership models. All meetings will be documented in a partner engagement log maintained by Cambridge Enterprise, and signed NDAs will be retained as formal verification records.

T4.3: Exploitation pathway decision and commercialisation plan - Synthesise the TEA outputs, IP status, prototype validation results, and consolidated partner feedback into a formally documented exploitation pathway decision by month 15. The selected pathway - licensing, spin-out, or joint development agreement - will be justified against explicit commercial and strategic criteria. A detailed commercialisation plan, including next-step milestones, resource requirements, and risk assessment, will be completed by month 18 and submitted as Deliverable D4.2, forming a key component of the PoC final report.

3.2 Team and Resources

Organisation	Country	Type	Role	Description
Principal Investigator, University of Cambridge	United Kingdom	Higher Education Institution	Coordinator	Principal Investigator and sole beneficiary

CATALYX-PoC is a single-PI ERC Proof of Concept action, as permitted and indeed anticipated by the ERC PoC instrument for cases in which the relevant innovation originates from a single principal investigator. The University of Cambridge provides the complete institutional infrastructure required for successful execution: state-of-the-art electrochemistry laboratories equipped for flow-cell electrolyser testing and online product analysis; Cambridge Enterprise, one of Europe's most active and experienced university technology transfer offices, providing IP management, FTO analysis coordination, spin-out formation capability, and direct industry engagement; and an established network of industry contacts in the chemical, energy, and cleantech sectors developed through the University's industrial liaison and innovation programmes. The concentration of all activities within a single, highly capable institution ensures efficient decision-making, clear accountability, and streamlined management of the limited PoC budget. Cambridge Enterprise's involvement from month 1 ensures that commercial and IP strategy considerations are integrated into technical decisions throughout the project, rather than being addressed only at the end.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	TEA process model completed	WP1	M5	Internal TEA model document reviewed by PI and Cambridge Enterprise
MS2	Provisional patent application filed	WP2	M3	Patent office filing receipt
MS3	Prototype electrode fabricated at 10 cm ² scale	WP3	M6	Performance data sheet confirming >80% FE for ethylene
MS4	TEA report and market benchmarking delivered	WP1	M9	Completed TEA report with sensitivity analysis
MS5	500-hour stability run completed	WP3	M15	Dataset report with chronoamperometry and GC records
MS6	PCT application filed	WP2	M12	PCT filing receipt
MS7	Exploitation pathway decision documented	WP4	M15	Board-level memo co-signed by PI and Cambridge Enterprise
MS8	Final commercialisation plan and PoC report	WP4	M18	Submitted final report and commercialisation plan document

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Techno-Economic Assessment	WP1	M9	Report	Confidential

	Report				
D2.1	Freedom-to-Operate Analysis Report	WP2	M4	Report	Confidential
D3.1	Prototype Performance Dataset (500 h)	WP3	M15	Data	Confidential
D4.1	Partner Engagement Summary and NDA Log	WP4	M15	Report	Confidential
D4.2	Final Commercialisation Plan	WP4	M18	Report	Confidential
D4.3	PoC Final Report	WP4	M18	Report	Public

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	Loss of selectivity or stability when the catalyst is fabricated at 10 cm ² electrode scale, compared to laboratory-scale results.	Medium	High	The scale-up strategy begins with systematic ink formulation and deposition optimisation (T3.1) before committing to long-duration testing. Alternative electrode architectures identified during the ERC project are available as fallback. TEA sensitivity analysis will quantify the minimum acceptable selectivity, allowing rational go/no-go decisions.
R2	Blocking prior art identified in the FTO analysis that narrows the scope of patentable claims.	Medium	Medium	FTO analysis is initiated in month 1 (T2.1), prior to the provisional filing deadline, allowing claim scope to be tailored to freedom-to-file positions. Cambridge Enterprise has experience navigating complex IP landscapes in catalysis. Trade secret protection for the synthesis process is maintained as a complementary IP strategy if patent scope is limited.
R3	Insufficient industrial partner interest to support a licensing deal or joint	Low	High	Partner engagement begins in month 6, once prototype data and TEA findings are

	development agreement within the PoC timeline.			available to support substantive technical discussions. A spin-out formation pathway via Cambridge Enterprise is maintained as an alternative route to commercial exploitation if direct industry licensing proves premature. The PI will leverage existing networks from the parent ERC project with industry contacts in the CCU sector.
R4	Electricity price volatility materially worsening the TEA economics and reducing investment attractiveness.	Medium	Medium	TEA model explicitly incorporates sensitivity to electricity price across a range of EUR 30–120/MWh, reflecting current and projected renewable electricity costs. The commercialisation narrative is framed around long-term renewable electricity cost trajectories and regulatory carbon pricing, both of which are favourable on a 5–10 year horizon relevant to industrial deployment decisions.
R5	Key personnel (postdoctoral researcher) leaving the project before completion.	Low	Medium	The PI holds sufficient expertise to supervise and partially execute the experimental programme. Recruitment of a replacement postdoc can be initiated rapidly through the University of Cambridge's established recruitment processes. Critical know-how is documented continuously through lab notebooks and internal reports to ensure continuity.